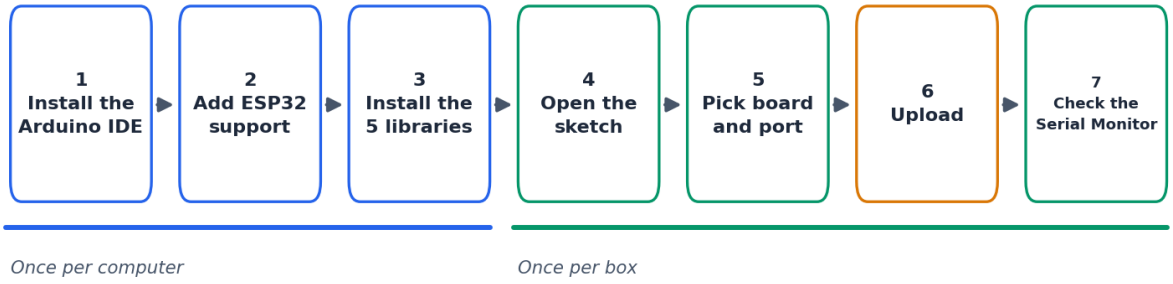
Installing the Firmware

Putting the Magic Boxes firmware onto an ESP32 board with the Arduino IDE

For: Technician / teacher • **Time needed:** About 30 min the first time, 2–3 min per box after that • **Hardware:** ESP32 with VEML7700, BME280, MPU-6500, SCD41

You only need this when building a new box, replacing a board, or updating to a newer firmware version. Boxes that already work do not need reflashing — unless their firmware is too old for an activity you plan to run (see the table below). A Romanian version of this guide is `FLASHING_GUIDE_RO.pdf`.



The first three steps are done once on your computer. The last four are repeated for each box.

About the pictures in this guide

The screenshots come from Arduino IDE 2.3.10 on Windows 11 and from an Android phone. Other versions look slightly different, but the menu and button names are the same. **Red** boxes and numbers show where to click, in order; **green** marks the right choice or a good result; a red cross ⊗ marks a look-alike to avoid.

Which firmware your activities need

The box prints its version at start-up (`FW:`, see section 7).

Activities	What the box needs
The Air We Breathe, Indoor Pollution, CO_2 and Photosynthesis, Air Quality, Greenhouse Effect — anything with CO_2	An SCD41 CO_2 sensor, **and** `serial_wifi` v2.2.0 or newer. A box still running the old USB-only firmware (`serial_only`) should be reflashed with `serial_wifi_v2_5_0`.
Light Is Everywhere, Solar Energy, Solar Panels — outdoors in sunlight	v2.2.5 or newer. Older firmware stops at about 3 800 lux, so every sunny reading looks the same. Above about 10 000 lux the absolute value is approximate until checked once against a reference lux meter; comparisons between readings are reliable.
Motion activities near fans or motors	v2.2.5 or newer filters out vibration that otherwise shows up as a fake slow wobble.
Everything else	Any version from v2.2.1 on. Older ones report the motion axes upside down.

The firmware supplied with this guide, `serial_wifi_v2_5_0`, covers all of them, and also works over a USB cable alone: a classroom without WiFi can use it unchanged. It is the only firmware for the box: the old USB-only `serial_only` is no longer supplied.

1. What you need

Item	Notes
ESP32 development board	The brain of the box.
USB cable	**It must be a data cable.** Many phone-charging cables carry power only, and a power-only cable is the most common reason a board never appears in the IDE.
A computer	Windows, macOS or Linux.
The sketch folders	Supplied next to this guide.

2. Install the Arduino IDE

Download it from `https://www.arduino.cc/en/software` and install it. Version 2.x is shown here; 1.8.x also works, and its menus have the same names.

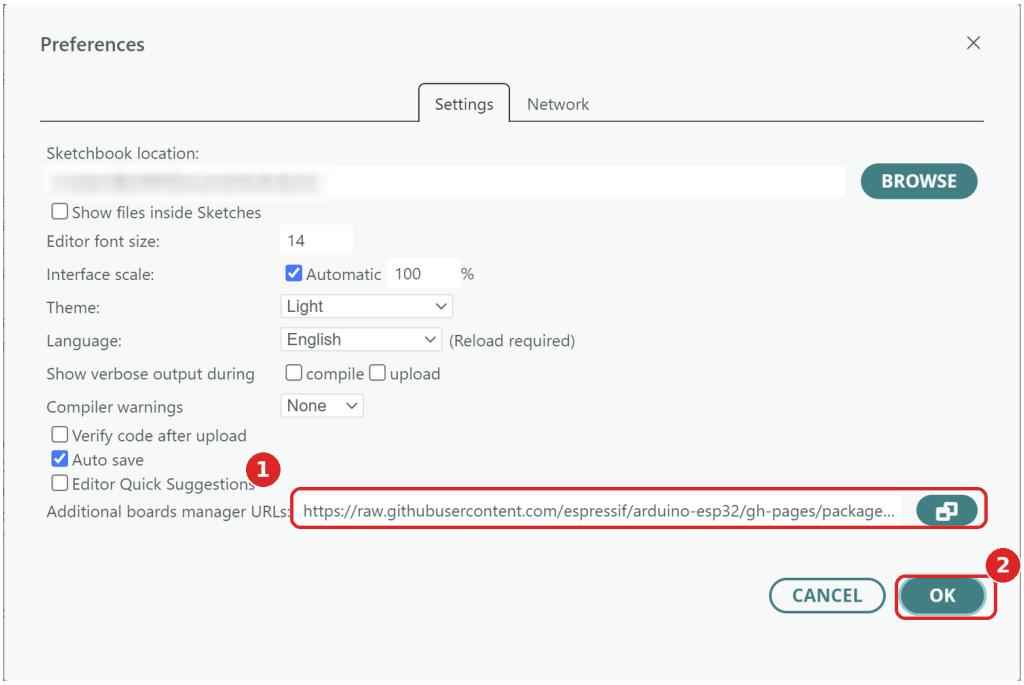
3. Add ESP32 support and the libraries

The IDE does not know about ESP32 boards until you tell it where to find them, and it does not know the sensors until the libraries are installed.

The board address

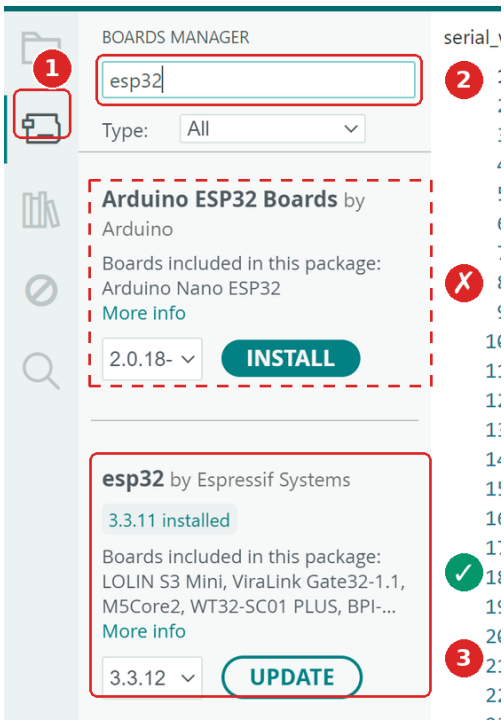
Open **File > Preferences**. In **Additional boards manager URLs** 1 paste this address, then press **OK** 2:

```
https://espressif.github.io/arduino-esp32/package_esp32_index.json
```



File > Preferences. The address in the picture is Espressif's older one, which also works; paste the one above.

The ESP32 board package



Boards Manager, searching for esp32.

Open the **Boards Manager**: the second icon on the left edge (1), or **Tools > Board > Boards Manager**. Type `esp32` in the search box (2).

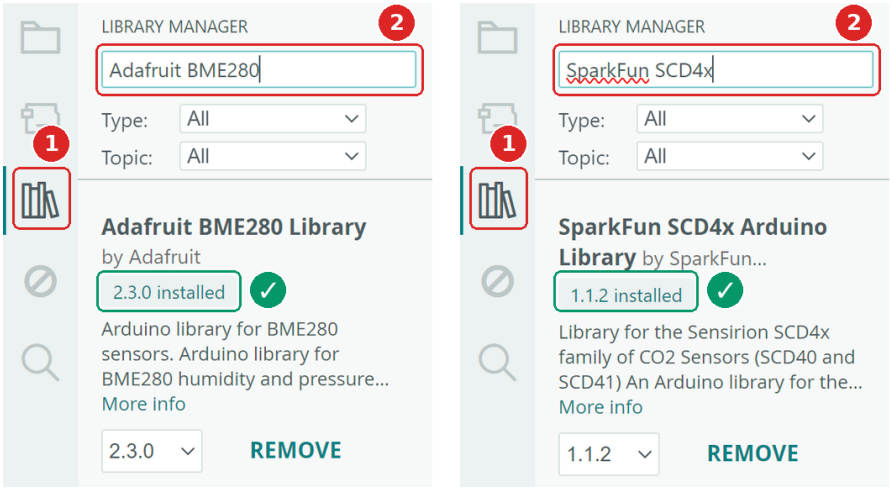
Install **esp32 by Espressif Systems** (3). It is a large download, so allow a few minutes. When it is done the card reads *installed*; in the picture it is already installed, so the button offers **UPDATE** instead of **INSTALL**.

Not the first result

The search shows **Arduino ESP32 Boards** by Arduino *first* (✗). That package only covers Arduino's own Nano ESP32 and cannot build this firmware. Scroll to **esp32 by Espressif Systems**.

The sensor libraries

Open the **Library Manager**: the third icon on the left edge (1), or **Tools > Manage Libraries**. Search for each of the eight libraries below by name (2) and press **INSTALL** on the card with exactly that name. Accept when the IDE offers to add a library's dependencies. An installed library shows *installed* on its card (✓).



Two of the eight, already installed: Adafruit BME280 Library and SparkFun SCD4x Arduino Library.

Library	Used for
Adafruit VEML7700	light sensor
Adafruit BME280 Library	temperature, humidity, pressure
Adafruit Unified Sensor	required by both Adafruit libraries above
ArduinoJson	building the data lines sent to the application
SparkFun SCD4x Arduino Library	CO_2 sensor
Adafruit seesaw Library	soil probe (moisture, temperature), optional
Adafruit SGP30 Sensor	SGP30 gas sensor (VOC), optional
Sensirion Gas Index Algorithm	VOC and NOx indices of the SGP41 gas sensor, optional

The WiFi, WebServer, DNSServer, Preferences and Wire libraries arrive with the ESP32 board package. Do not install them separately.

The last three rows serve sensors that most boxes do not have. Install them all the same: the sketch names every library, so it will not compile without them, whatever is plugged into the box.

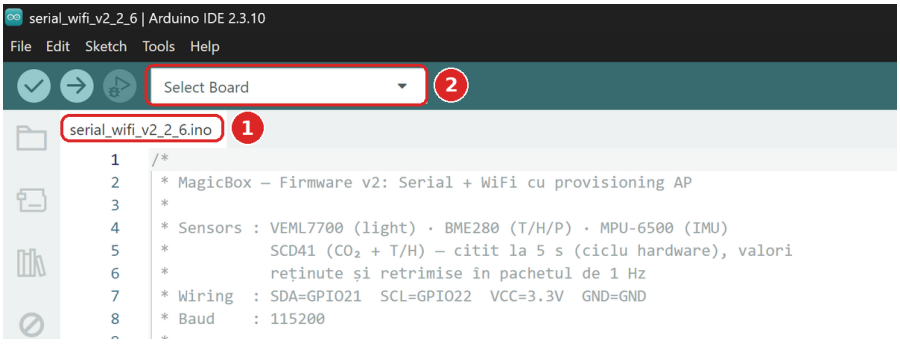
The particle sensor library (ZIP file)

The BMV080 particle sensor needs **DFRobot_BMV080**, which is not in the Library Manager. Open `https://github.com/DFRobot/DFRobot_BMV080`, press **Code > Download ZIP** and keep the file as it is, without unpacking it. In the IDE choose **Sketch > Include Library > Add .ZIP Library** and select that file. Once added, **DFRobot_BMV080** appears under **Sketch > Include Library**. This library, too, is needed to compile the sketch for every box, with or without a particle sensor.

4. Open the sketch

The firmware is `serial_wifi_v2_5_0`. It drives all four sensors, also reads the soil, gas and particle sensors when they are plugged in, and speaks to the application over both USB and WiFi.

Open it with **File > Open** and select `serial_wifi_v2_5_0.ino` inside the folder of the same name. Arduino requires a sketch and its folder to share a name, so keep the folder exactly as supplied.

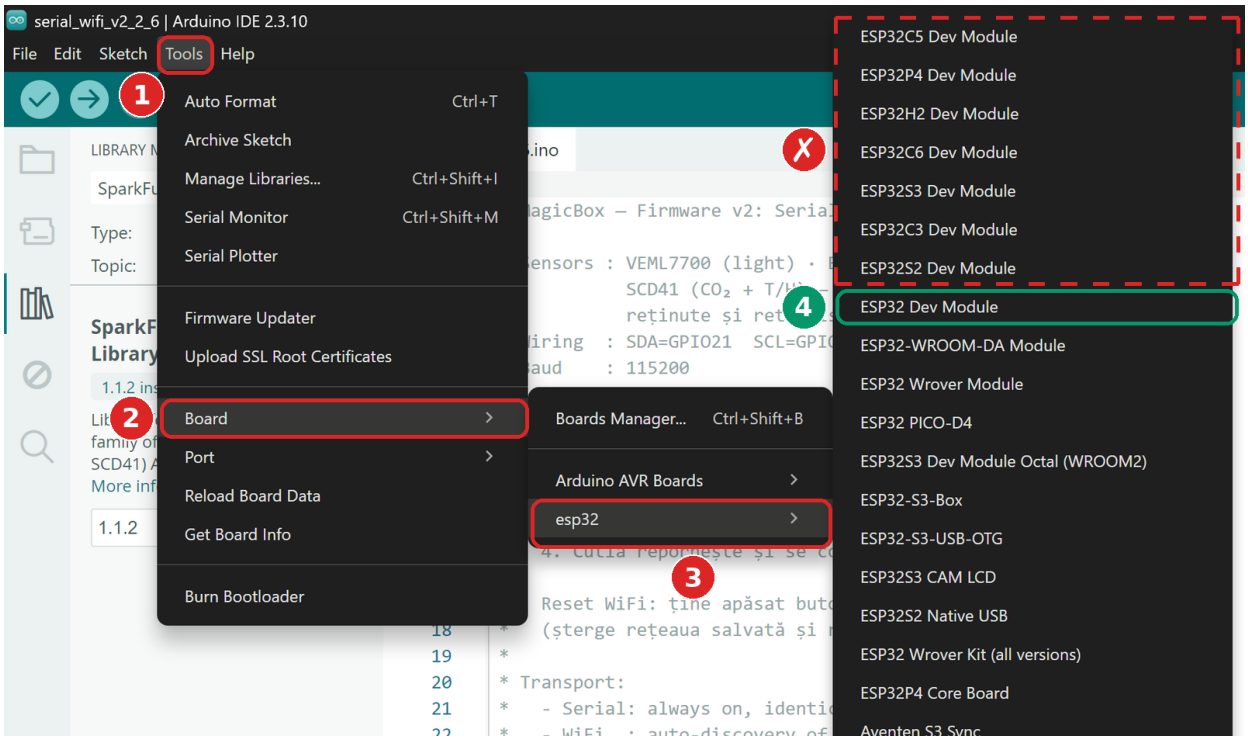


The sketch is open when its name shows on the tab 1. Select Board 2 *means no board is chosen yet: that is the next step.*

5. Select the board and the port

Plug the board in with the USB cable first, so that its port exists.

1. **Tools** 1 **> Board** 2 **> esp32** 3 **> ESP32 Dev Module** 4, which suits the boards used in the Magic Boxes.



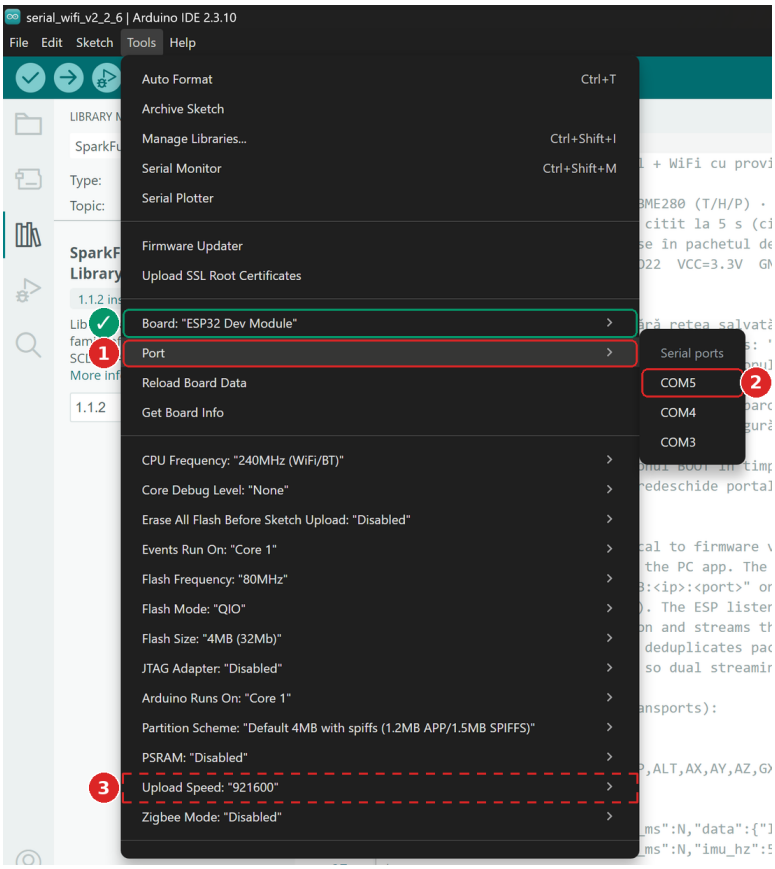
Tools > Board > esp32. ESP32 Dev Module is the eighth entry, below seven look-alikes.

ESP32 Dev Module — exactly

The list starts with **ESP32C5, ESP32P4, ESP32H2, ESP32C6, ESP32S3, ESP32C3** and **ESP32S2 Dev Module** ✗. Those are other chips: with one of them selected the upload fails, or the firmware does not run. Pick the entry that reads **ESP32 Dev Module** and nothing more.

2. **Tools > Port** 1 > the port that appears when the board is plugged in 2: `COM5` or similar on Windows, `/dev/cu.usbserial-…` on macOS, `/dev/ttyUSB0` on Linux. If several are listed, unplug the board, look again, and take the one that disappeared.

3. Leave everything else at its default. The menu now reads **Board: "ESP32 Dev Module"** ✓. If uploads fail repeatedly, set **Upload Speed** 3 to 115200.



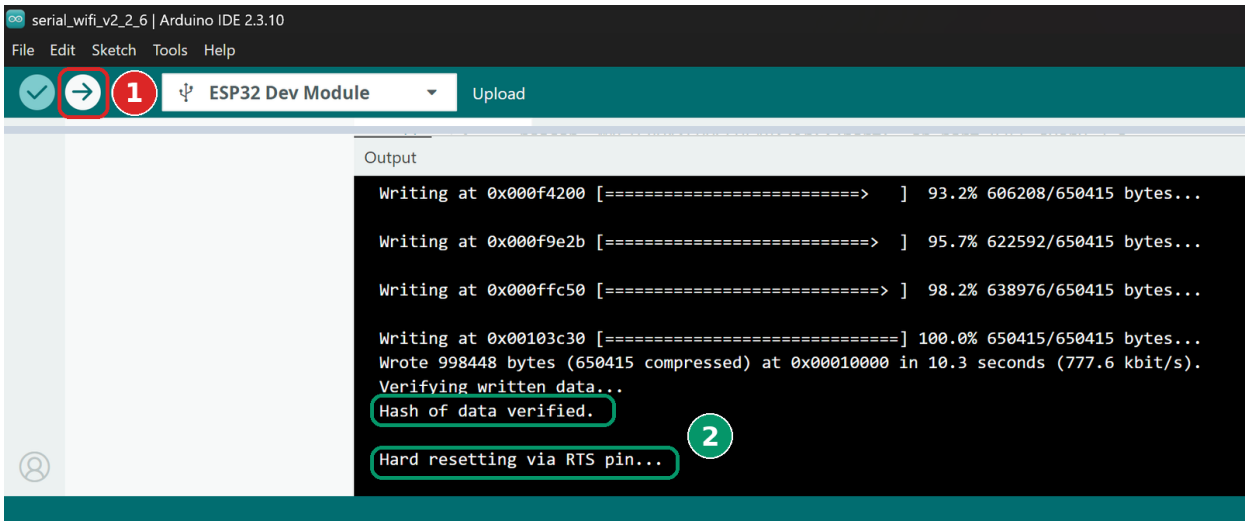
Tools > Port, with the board already chosen.

No port in the menu

The computer is missing the USB-to-serial driver for the chip on your board. Install the **CP210x** driver from Silicon Labs, or the **CH340** driver, depending on which chip the board carries — its documentation will say. Unplug, replug, and look again. If the port still does not appear, try another cable before anything else.

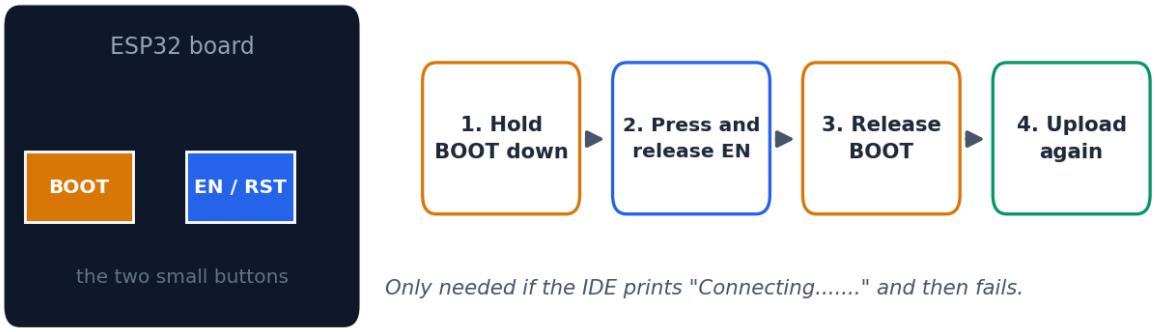
6. Upload

Press **Upload**, the right-pointing arrow 1. The first compilation of an ESP32 sketch takes a couple of minutes; later ones are much quicker. The upload has worked when the Output panel ends with **Hash of data verified** and **Hard resetting via RTS pin...** 2.



A finished upload. The sketch text in the middle of the window is left out.

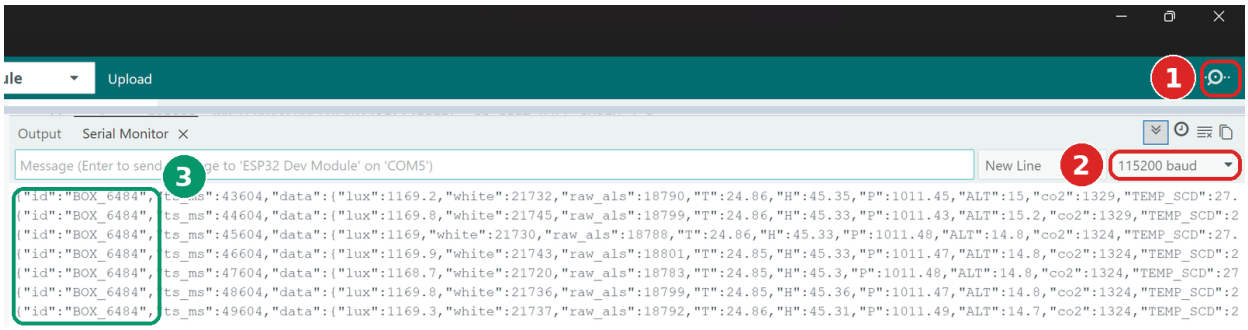
Some boards need help entering upload mode. If the Output panel shows `Connecting…___` and then fails:



The button sequence that puts a stubborn board into upload mode.

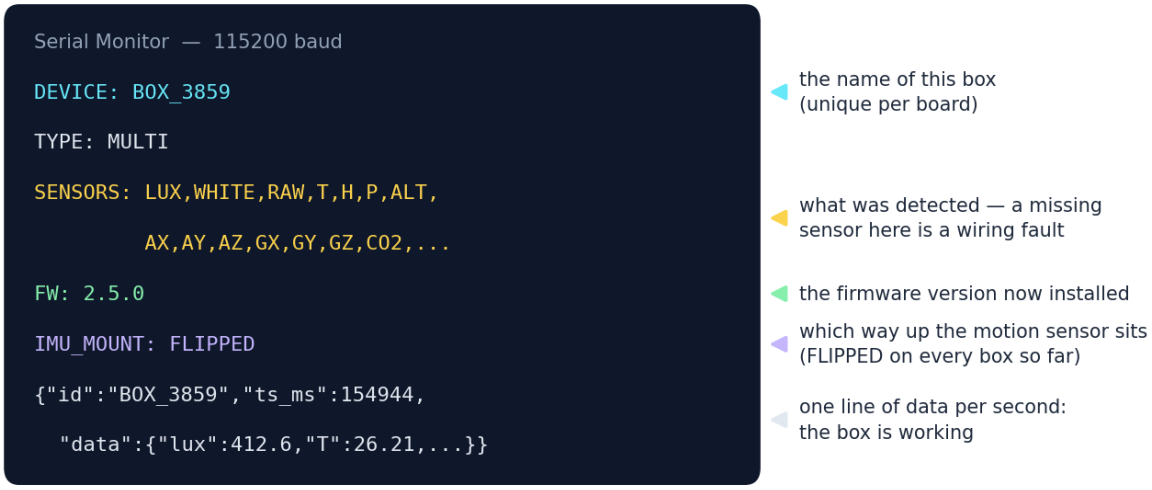
7. Check that it worked

Open the **Serial Monitor**: the icon at the top right 1, or **Tools > Serial Monitor**. Set the speed to **115200 baud** 2. A data line a second should appear, each starting with the box's name 3.



The Serial Monitor at 115200 baud, with data from box BOX_6484.

The box introduces itself only in the first second after a restart, so those lines have usually scrolled past. To see them, press the board's **EN** (or **RST**) button with the Serial Monitor open:



A healthy box, and what each line tells you.

Close the Serial Monitor before starting the Magic Boxes application: only one program can hold the port at a time.

Boxes with the motion sensor the right way up (new board)

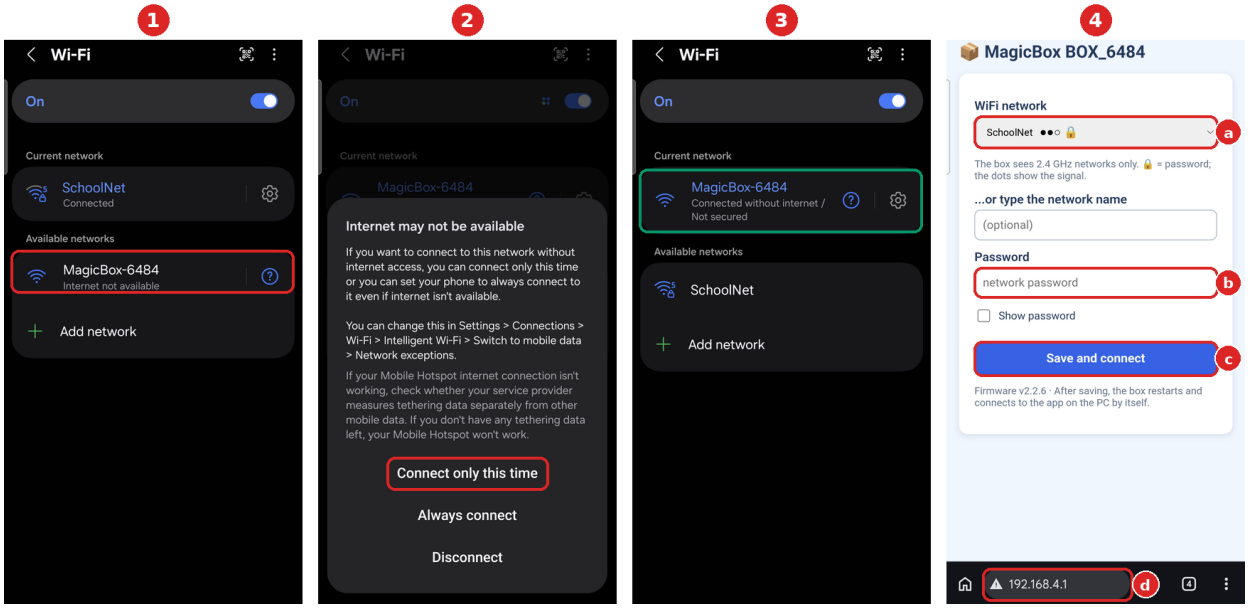
On every box built so far the motion sensor sits upside down on the board. The firmware turns its readings over, so a box lying flat reads about +1 g on acceleration Z, and the start-up line `IMU_MOUNT: FLIPPED` says so. A newer board has the sensor the right way up. With the same firmware such a box would read about −1 g lying flat, so it is set once to `NORMAL`. After flashing a box, check it once:

1. Lay the box flat on the table, connect it and open **Live Data** in the application.
2. Look at acceleration Z. About +1 g: nothing to do. About −1 g: the sensor sits the right way up, so tell the box.
3. Either in the application: **Settings > Box names**, tick **Motion sensor upside down** for that box while it is connected. Or in the Serial Monitor (115200 baud): type `IMU_MOUNT NORMAL` in the message box at the top and press Enter. The box answers `[IMU] Sensor mounting saved: NORMAL (readings unchanged)` and `IMU_MOUNT: NORMAL`.

The setting is kept in the box itself: it counts on every computer, and it survives switching off, reflashing and the BOOT-button WiFi reset. `IMU_MOUNT FLIPPED` sets it back. `INFO` prints the start-up lines again, so you can see a box's setting without restarting it. Upper or lower case does not matter.

8. Connect the box to WiFi

The network is configured on the box itself, so the sketch never contains a password. On first start, or whenever the saved network is unreachable, the box creates an open WiFi network called **MagicBox-XXXX**, where XXXX matches the end of the device name (**CutiaMagica-XXXX** on firmware older than 2.2.6).



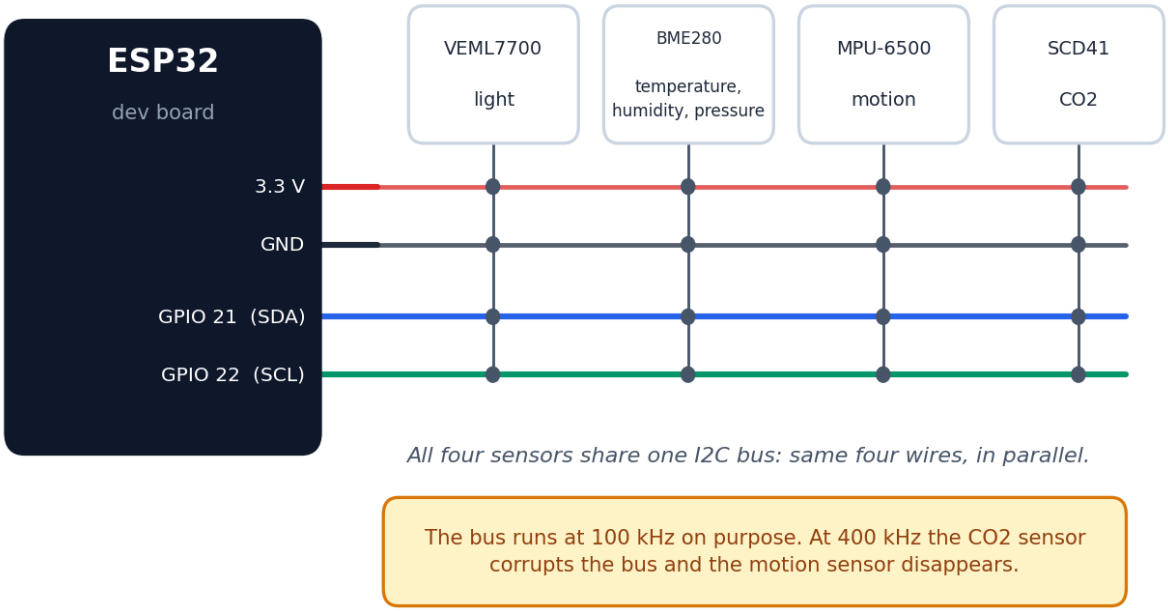
Joining the box from a Samsung phone. The steps are numbered as in the list below.

1. In the phone's WiFi settings, tap **MagicBox-XXXX**. It is listed as *Internet not available*; that is expected, the box has no internet.
2. The phone warns *Internet may not be available*. Choose **Connect only this time**. *Always connect* also works, but then the phone keeps returning to the box later.
3. The phone now shows *Connected without internet*.
4. Open the setup page. On some phones it opens by itself; on Samsung phones it does not, so open the browser and type **http://192.168.4.1** in the address bar (d). The page is in English:
   (a) under **WiFi network**, choose the classroom network. The list shows **2.4 GHz networks only**, the only kind the box can use; a hidden network can be typed under *…or type the network name*;
   (b) type its password under **Password**;
   (c) press **Save and connect**.

The box restarts, joins that network and from then on reconnects by itself. While its setup network is up and nobody is on the page, it tries the saved network again every 2 minutes. The phone can then return to its usual WiFi.

The application announces itself on the network every three seconds and the box connects back to it, so no addresses are typed anywhere. USB keeps working at the same time and the application ignores the duplicate data, so a cable can be unplugged mid-lesson without losing the session.

9. Wiring



Four wires, shared by all four sensors.

10. When something goes wrong

Symptom	Likely cause and remedy
No port in the Tools menu	Power-only USB cable, or a missing CP210x/CH340 driver.
`Connecting…` then a timeout	Use the button sequence above; try a different USB port or a shorter cable.
Compiling stops with `No such file or directory` after a `.h` name	That library is missing. Install it ("The sensor libraries" above); every one is needed, even for a box without that sensor.
Compiles, but the Serial Monitor stays empty	Wrong speed — it must be 115200 baud.
A sensor missing from `SENSORS:`	A wiring fault, not a firmware one. Check that device's four connections, and that it sits on 3.3 V rather than 5 V.
Values appear, then stop after a while	Update to the current firmware; older versions could be frozen by a stalled sensor.
Light stays at about 3 775 lux in bright light	The box runs firmware older than v2.2.5. Reflash it.
No CO_2 values although the sensor is fitted	The box has the old USB-only firmware (`serial_only`): reflash it with `serial_wifi_v2_5_0`.
Lying flat, the box reads about −1 g on acceleration Z	The board has the motion sensor the right way up. Set it once to `NORMAL`, see "Boxes with the motion sensor the right way up" above.
The application does not see the box	Close the Serial Monitor, and check that box and laptop are on the same network.
Over WiFi the box never connects, though it is on the right network	The laptop's firewall is blocking the application. Allow it on private networks (TCP port 8765, UDP port 8766). The box retries on its own, at most a minute apart; restarting the application makes it try at once.

Handling

Work on a non-conductive surface and unplug the board before changing any wiring. The sensors run at 3.3 V — connecting one to the 5 V pin can destroy it. Keep boards away from liquids, and hold them by the edges: the pins bend easily and a bent pin is the usual cause of a sensor that has disappeared.

11. Which version am I running?

The box prints `FW:` at boot, and `CHANGELOG.txt` in this folder lists what changed in each version and when. The current firmware is **v2.5.0**. It adds the optional soil, gas and particle sensors; a box without them sends exactly what it sent with v2.3.0. Boxes on v2.2.6 or v2.3.0 need updating only when they get one of these sensors. Version 2.3.0 itself only added the sensor setting for the new board.